Nifty Commodities Benchmark Return Prediction

Machine Learning with Business Data mini project

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Member	SRN
Darshan Govekar	PES1UG24AM355
Dhanraj Rathod	PES1UG24AM359
Shridhar Umesh Golipalle	PES1UG25AM814

Abstract

This project predicts the next observed trading session return of the Nifty Commodities price index from information available after the current close. We compare XGBoost, Keras, fastai, PyTorch TabNet and Lightning Flash using the same 23 numeric features and chronological partitions. Official NSE Indices data provide 3,644 complete OHLC observations from 27 February 2012 to 9 November 2026. Three expanding folds within the training era use separate inner stopping periods, followed by a 2022 to 2023 development validation period and a frozen test period beginning in 2024.

The lowest validation RMSE selects fastai. Its test RMSE is 1.1821 percentage points, MAE is 0.8329, and R2 is -0.0025. The later test shows little forecasting advantage over simple predictions. A weighted average of two models is an optional extension, and TreeSHAP explains the XGBoost model. These results concern forecast errors; they do not establish trading profits.

Project scope

We study one question: can the index price history help predict its next session return? The workflow is data preparation, EDA, feature creation, training the five required frameworks, and comparing their results. The report explains each step; the presentation follows the same order.

Introduction and Data

Forecasting objective

The Nifty Commodities index represents listed Indian companies in commodity-related industries. The official October 2026 factsheet describes 30 companies. It is an equity price index, rather than a basket of physical commodity spot prices. The index launched on 7 September 2011 and has a base date of 1 January 2004. The modeling sample starts after launch. [1]

At the official close of session t, the model estimates the percentage return from that close to the close of the next observed session. The target is y(t) = 100 x [C(t+1) / C(t) - 1]. A prediction of 0.4 means a 0.4% index return. RMSE and MAE therefore use percentage points. The price series excludes reinvested dividends and must not be described as a total return benchmark.

Official source and audit trail

The downloader submits calendar-year requests to the historical endpoint linked by NSE Indices. All 17 JSON responses are retained with request payloads, retrieval times and SHA256 checksums in data/raw/provenance.json. The frozen request ends on 10 November 2026. The latest returned session is 9 November because 10 November is an exchange trading holiday. [2]

Quality item	Observed treatment
Downloaded sessions	4,182 from 4 Jan 2010 to 9 Nov 2026
Missing initial OHL	538 sessions through 24 Feb 2012 excluded
Complete modeling era	3,644 sessions from 27 Feb 2012
Interior OHLC omissions	None in the retained continuous era
Duplicates and invalid prices	No duplicates, nonpositive prices or inconsistent OHLC retained
Market extremes	Preserved without winsorizing or clipping
Holidays and weekends	No rows fabricated and no prices filled

Excluding a contiguous initial period avoids creating invented intraday ranges. Returns still use adjacent observed market sessions. The downloader also retains the longer close series, so the exclusion is inspectable rather than hidden.

Exploratory Data Analysis

Training data drive exploratory interpretation. Full-history plots show context and partition boundaries, while feature correlations and distribution summaries use training observations only. The final test labels do not determine features, model settings, blend weights or model selection.

Training target statistic	Value
Mean daily return	0.0476%
Standard deviation	1.2769 pp
Minimum and maximum	-12.2484% to 7.6024%
Skewness	-0.6716
Excess kurtosis	7.6595
Positive return share	53.28%

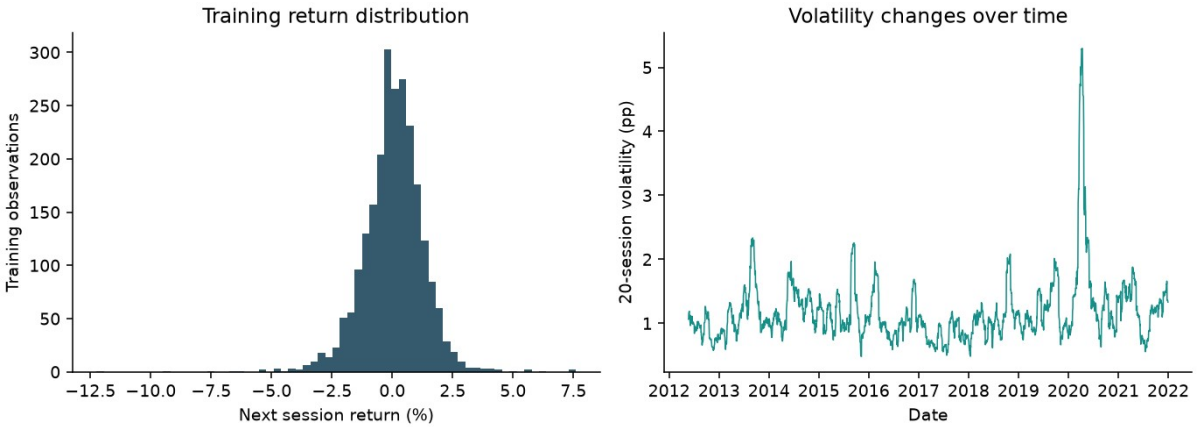


Figure 1 Training returns and rolling volatility

The distribution and rolling volatility expose the scale of daily forecasting error and changes in risk. Large observed returns remain in evaluation. The rolling estimates are trailing calculations, so a future shock cannot change a past feature. A large excess kurtosis indicates tail behavior that a Gaussian error assumption would miss.

History and Causal Features

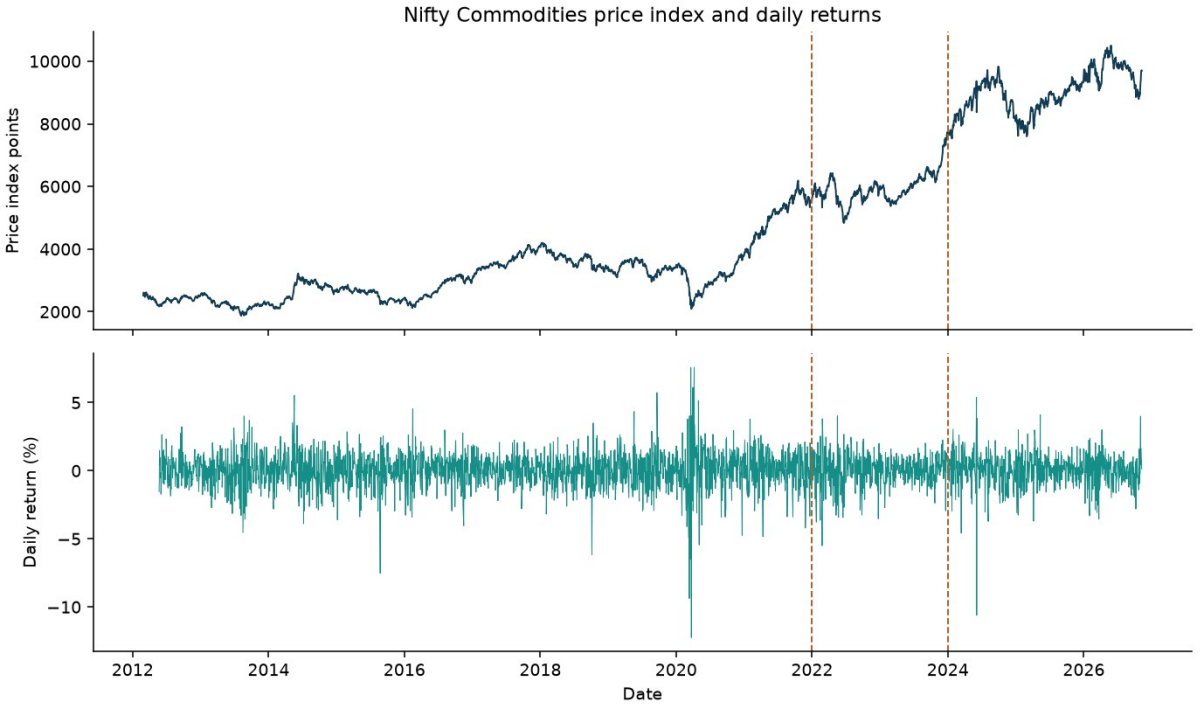


Figure 2 Historical context with development and test boundaries

Feature group	Definitions available at close t
Short returns	Current close return and lagged returns at 1, 2, 3 and 5 sessions
Momentum	Close changes over 5, 10 and 20 observed sessions
Rolling statistics	Return mean and sample volatility over trailing 5 and 20 sessions
Price position	Close relative to trailing 5 and 20 session means
Intraday conditions	Close/open return, high-low range/close, open/previous close gap
Oscillator and drawdown	14-session simple-average RSI and close relative to 60-session maximum
Known calendar	Weekday uses a seven-day cycle; month uses a twelve-month cycle

There are 23 input columns. The 60-session drawdown creates a 59-row warm-up and the final unlabeled row is removed. Features use neither centered windows nor backward filling. The exact feature order is saved in models/features.json. Correlated features can share explanatory credit, so a large attribution is not evidence of an independent economic cause.

The source includes 18 special weekend sessions. The weekday encoding keeps all seven days distinct. This avoids making Saturday share Monday values or Sunday share Tuesday values. Weekend sessions remain real observations and the target is still the next observed session.

Methodology and Validation

Partition	Feature dates	Rows
train	2012-05-22 to 2021-12-30	2380
validation	2022-01-03 to 2023-12-28	493
test	2024-01-01 to 2026-11-08	709

The training cutoff is 31 December 2021 and the development validation cutoff is 31 December 2023. Boundary purging removes a row whenever its outcome reaches the next partition first forecast date. The test starts in 2024 and remains untouched during fitting, stopping and selection. StandardScaler learns the mean and population standard deviation from training features only. Each later partition uses that same transformation. XGBoost receives the same transformed inputs for a consistent comparison, although trees do not require scaling.

Expanding cross validation

As an additional historical check, we repeat training and scoring in three earlier periods. Each check trains on the past and scores the following 252 sessions. A separate 126-session block controls when training stops. A boundary row is removed so its target cannot reach the next block. The scaler learns only from the fitting block. This is expanding cross validation, following Unit 1 page 81 and Unit 2 page 59. [3, 4, 5]

All five frameworks use the same fold indices, features, target, CPU setting and seed 42. Architecture settings are intentionally compact. The project does not claim an exhaustive search or identical parameter counts across different model families. Final fitting uses the full pre-2022 training block and monitors the 2022 to 2023 validation block. No post-selection refit changes the frozen test predictions.

Metrics and selection

RMSE = sqrt(mean((y - prediction)^2)) gives more weight to large errors. MAE = mean(abs(y - prediction)) is the average absolute error. Lower is better for both. R2 = 1 - sum((y - prediction)^2) / sum((y - evaluation mean)^2); higher is better and negative values are possible. Direction accuracy checks whether the forecast sign matches the actual sign, treating zero as nonpositive. It is an extra diagnostic derived from regression.

The lowest development validation RMSE selects the recommendation from all models and baselines. The report displays every frozen test score for comparison. A model that happens to score best on the test is descriptive and does not replace the validation-selected recommendation.

Framework Implementations

Framework	Actual fitted model and settings
XGBoost	Squared-error boosted trees, depth 2, up to 160 trees, learning rate 0.03, subsample 0.9, column sample 0.9, L2 5, stopping patience 20.
Keras	Keras 3 with PyTorch backend. Dense 64 and 32 ReLU layers, dropout 0.1, L2 0.001, linear output and Adam at 0.001.
fastai	TabularPandas and tabular_learner. Dense 64 and 32 layers, dropout 0.1, no batch normalization, Adam at 0.001 and weight decay 0.001.
PyTorch TabNet	TabNetRegressor with n_d = n_a = 8, three attentive steps, gamma 1.3, sparse penalty 0.001 and Adam at 0.01.
Lightning Flash	A real Flash Task and Flash Trainer around a PyTorch 64 and 32 ReLU MLP, dropout 0.1 and Adam at 0.001 with weight decay 0.001.

Deep models use at most 40 epochs and stopping patience 5. The dense models use batches of 64. TabNet uses batches of 256 with virtual batches of 64. MSE is the regression objective. Best monitored weights are restored before predictions. Model metadata save the actual stopping epoch, package versions, histories and elapsed training-plus-prediction time. [6, 7, 8, 9]

Lightning Flash uses its real Task and Trainer APIs with a small regression network. Its older release needs pinned dependencies. The exact compatibility settings are recorded in docs/REPRODUCIBILITY.md and the environment lock. This keeps setup details separate from the forecasting method.

Simple baselines and optional blend

Zero return predicts no change. Training mean predicts the average training return every day. Ridge is a regularized linear regression. These baselines help judge whether a complex model adds value. They use no test outcomes for fitting.

The optional blend is a weighted average of XGBoost and a deep model. Validation chooses fastai as the deep component and checks eleven weights. It selects 0% XGBoost and 100% fastai. A zero XGBoost weight means the blend is simply the deep model; it provides no extra gain. The main recommendation remains the single candidate with the lowest validation RMSE.

Development Results

Outer fold cross validation

Model	RMSE mean and sample SD	MAE mean and sample SD
xgboost	1.4966 +/- 0.4564	1.0719 +/- 0.1974
keras	1.5183 +/- 0.5090	1.0799 +/- 0.2274
fastai	1.4954 +/- 0.4642	1.0665 +/- 0.2043
tabnet	1.5205 +/- 0.4503	1.0819 +/- 0.1963
flash	1.4930 +/- 0.4619	1.0683 +/- 0.2037

These three outer-fold scores measure sensitivity to historical eras. Their sample standard deviation describes variation across three periods and is not a confidence interval. The same stopping protocol precedes every outer score. Complete per-fold dates, row counts and signed R2 values are saved in results/cv_metrics.csv.

Development validation scores

Model	RMSE pp	MAE pp	R2	Direction
zero_return	1.1234	0.8360	-0.0038	46.65%
training_mean	1.1214	0.8333	-0.0004	53.35%
ridge	1.1282	0.8442	-0.0124	51.72%
xgboost	1.1215	0.8329	-0.0004	53.35%
keras	1.1172	0.8319	0.0072	54.56%
fastai	1.1096	0.8250	0.0207	53.55%
tabnet	1.1201	0.8332	0.0020	53.35%
flash	1.1124	0.8298	0.0156	53.55%
blend	1.1096	0.8250	0.0207	53.55%

The selection rule chooses fastai, with validation RMSE 1.1096 pp. The optional blend ties it because its selected weight reduces to the same model. Validation also controls when training stops, so these scores may be optimistic. The later test checks performance after these choices are fixed.

Frozen Test Results

Model	RMSE pp	MAE pp	R2	Direction
zero_return	1.1813	0.8428	-0.0011	44.29%
training_mean	1.1807	0.8385	-0.0001	55.71%
ridge	1.1887	0.8482	-0.0137	48.52%
xgboost	1.1786	0.8379	0.0034	55.99%
keras	1.1667	0.8485	0.0235	53.03%
fastai	1.1821	0.8329	-0.0025	55.01%
tabnet	1.2013	0.8306	-0.0354	55.01%
flash	1.1741	0.8329	0.0110	55.29%
blend	1.1821	0.8329	-0.0025	55.01%

The validation-selected fastai records test RMSE 1.1821 pp, MAE 0.8329 pp, R2 -0.0025 and direction accuracy 55.01%. The lowest observed test RMSE belongs to keras. That retrospective comparison does not change the earlier selection.

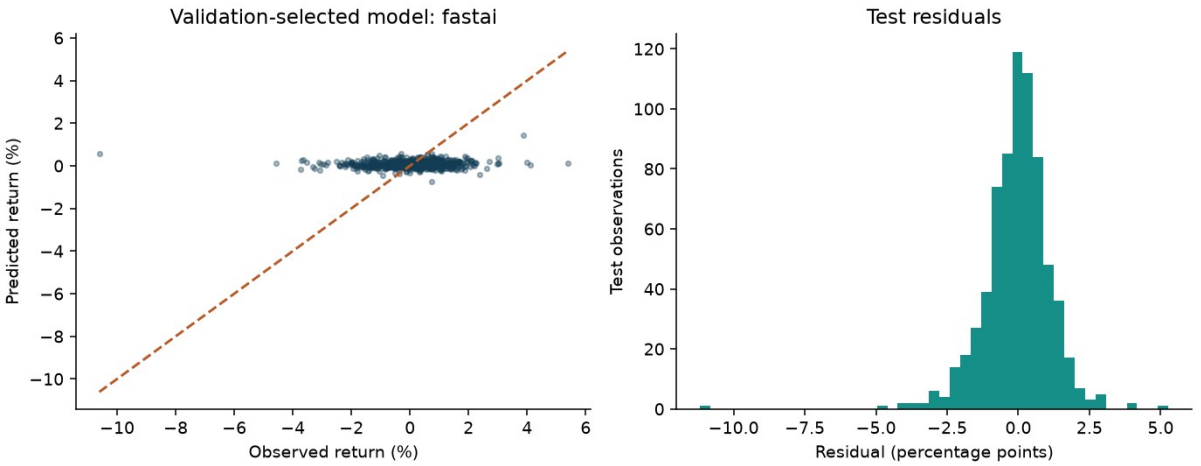


Figure 3 Observed versus predicted return and residuals of the selected model

A negative R2 means that the model performs worse than predicting the mean of the evaluation outcomes in the R2 denominator. That denominator is a retrospective reference, not a deployable baseline. Compare the actual zero-return and training-mean predictors separately. Good level prediction does not imply good return prediction, so every metric here evaluates the return target.

Interpretability and Practical Comparison

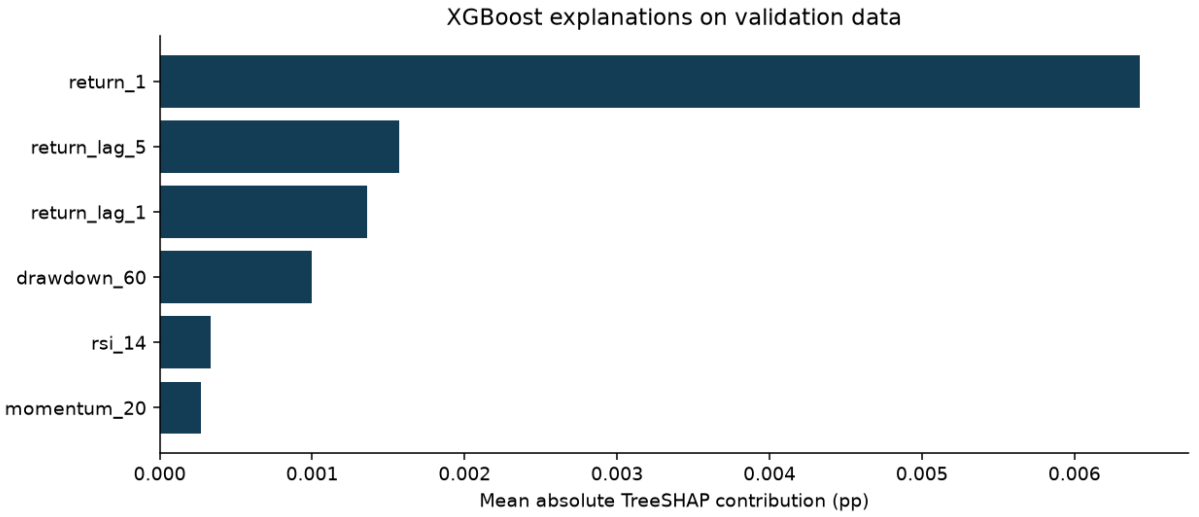


Figure 4 Validation TreeSHAP attributions for the stopped XGBoost model

Native XGBoost TreeSHAP decomposes each prediction into a bias term and feature contributions. The computation uses only trees through the best stopping iteration. The maximum numerical additivity error is 7.45e-09 pp. On validation data, the three largest mean absolute attributions are return_1, return_lag_5, return_lag_1. These describe the fitted forecast, not causal drivers or investment value. [10]

Framework	Simplicity	Transparency	Efficacy
XGBoost	Short tabular API	TreeSHAP used here	Test RMSE 1.1786 pp
Keras	Readable layer API	Needs extra attribution	Test RMSE 1.1667 pp
fastai	High-level learner	Needs extra attribution	Test RMSE 1.1821 pp
TabNet	More settings	Feature masks available	Test RMSE 1.2013 pp
Flash	Legacy setup work	Needs extra attribution	Test RMSE 1.1741 pp

Simplicity and transparency are reasoned qualitative comparisons, not invented numeric ratings. Efficacy is the observed RMSE, MAE, R2 and direction accuracy on identical dates. The simple baseline remains preferable when added complexity fails to improve the chosen validation criterion.

Conclusion and Reproducibility

Validation selects fastai. On the later test it records RMSE 1.1821 pp, compared with 1.1813 pp for zero return. Its R2 is -0.0025 and direction accuracy is 55.01%, compared with 55.71% for the training-mean baseline. This is very weak predictive skill. The lowest test RMSE belongs to keras, but that observation does not change the earlier validation choice. The optional blend adds no gain in this run.

Limitations

One index and one final test era limit generalization. Inputs contain only the index own OHLC history and known calendar values. They omit macroeconomic releases, exchange rates, commodity futures, news and fundamentals. The experiment uses one seed, so it does not estimate neural initialization uncertainty. Historical data can be revised by the provider.

The final model remains frozen at its original training cutoff, which makes evaluation auditable but can become stale. Rolling retraining and additional independent eras would be the next experiment. They must be specified before looking at new outcomes. A signal computed after the official close cannot assume execution at that same closing price. No trading return, Sharpe ratio or profit claim is made.

Files and commands

Start with README.md for the short explanation and file map. Run .\run_demo.ps1 for one saved historical prediction. Run .\run_all.ps1 to retrain, test, rebuild the notebook, Word report, PDF and PowerPoint, and refresh the submission ZIP. The full authoring command uses the prepared Codex runtime; on a computer with only Python, .\run_all.ps1 -ResultsOnly rebuilds the experiment and notebook.

Each full training run fits every model again. There is no prediction-cache shortcut. Downloading is an explicit option, so normal runs use the included frozen data. The code and deliverables are in the private GitHub repository https://github.com/Darshansg76/Team12-Nifty-Commodities-Return-Prediction on main. Other people need repository access to open it.

Connection to the four units

Course material	Application
Unit 1 pp 81 and 145 to 163	Time-based forecasting and regression evaluation with signed R2
Unit 2 pp 7 to 8 and 47 to 59	Train-only preprocessing, common folds and untouched final test
Deep Learning 1.pdf pp 6 and 25 to 33	Two hidden layers, fair features, stopping and best checkpoints
Interpretability 3.pdf to 6.pdf	Local explanations, SHAP and limits of feature attribution

References and Source Notes

[1] NSE Indices Limited. Nifty Commodities factsheet, 30 October 2026. https://www.niftyindices.com/Factsheet/ind_nifty_commodities.pdf. Local copy retained in references.

[2] NSE Indices. Historical data and index page. https://www.niftyindices.com/reports/historical-data and https://www.niftyindices.com/indices/equity/thematic-indices/nifty-commodities. Official trading holiday calendar: https://www.nseindia.com/resources/exchange-communication-holidays. Retrieved 11 November 2026. Requests and checksums are in data/raw/provenance.json.

[3] Supplied unit 1 merged.pdf. Time-based split p 81. Regression metrics pp 145 to 163. Page references use physical PDF positions.

[4] Supplied unit 2 merged.pdf. Leakage pp 7 to 8 and 25. Common evaluation and time-series validation pp 47 and 59. Detailed mapping is in references/unit_1_2_mapping.md.

[5] Scikit-learn. TimeSeriesSplit API. https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html.

[6] Chen, T. and Guestrin, C. 2016. XGBoost A Scalable Tree Boosting System. https://arxiv.org/abs/1603.02754.

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[9] Lightning Flash. Task API. https://lightning-flash.readthedocs.io/en/stable/api/generated/flash.core.model.Task.html. Supplied Deep_Learning_with_Tabular_Business_Data/1.pdf and 3.pdf. Page mapping in references/unit_3_4_mapping.md.

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